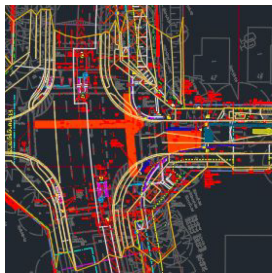


Augmented Reality for Tabletop and In-Situ Visualization of Road Construction Drawings to Support Engineers

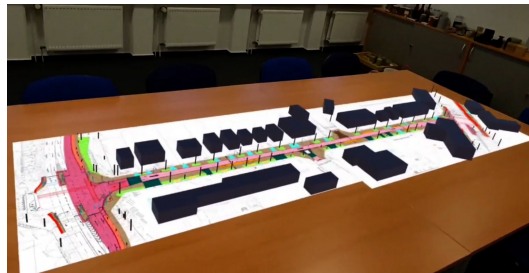
Jarne Münster
Human-Computer Interaction
University of Hamburg
Hamburg, Germany
jarne.muenster@studium.uni-hamburg.de

Julia Hertel
Human-Computer Interaction
University of Hamburg
Hamburg, Germany
julia.hertel@uni-hamburg.de

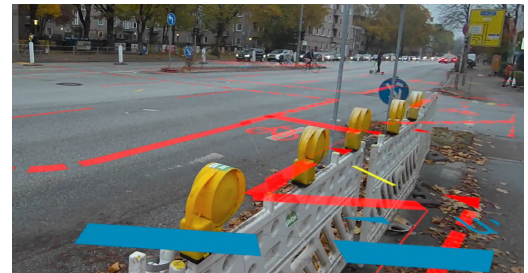
Frank Steinicke
Human-Computer Interaction
University of Hamburg
Hamburg, Germany
frank.steinicke@uni-hamburg.de



(a) Original 2D road construction drawing



(b) Tabletop visualization



(c) In-situ visualization

Figure 1: AR visualization of road constructions drawings, aiming to increase spatial understanding during the planning phase.

Abstract

Planning road construction projects is a complex process which requires several aspects to be taken into account in order to ensure a successful construction and subsequently smooth and safe traffic flow. During the planning phase, engineers create construction drawings that are typically viewed on 2D desktop interfaces and must be mentally transferred to 3D to be validated. This work aims to investigate whether Augmented Reality (AR) has potential to support engineers during the planning phase by displaying construction drawings either as a 3D tabletop for the use in offices, or as an in-situ visualization for the use on the planned construction site. To assess the potential, a prototype is developed and evaluated in a focus group field study with engineers. The results suggest that engineers see potential for AR, both for the use in offices and on-site. Further, additional feature wishes and current challenges are presented.

CCS Concepts

• **Human-centered computing** → **Field studies; Mixed / augmented reality.**

Keywords

Augmented Reality, Visualization, Field Study, Infrastructure, Road Planning

ACM Reference Format:

Jarne Münster, Julia Hertel, and Frank Steinicke. 2026. Augmented Reality for Tabletop and In-Situ Visualization of Road Construction Drawings to Support Engineers. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3772363.3799371>

1 Introduction

Road constructions, encompassing the planning, creation and maintenance of traffic ways, is a complex process, where aspects like traffic flow, safety, or drainage have to be considered simultaneously [7]. During the planning phase, engineers create construction drawings, i.e., complex plans including all relevant elements such as lanes, traffic lights and surrounding buildings, which are viewed and edited on traditional 2D desktop interfaces. Any issues which could lead to problems during the construction phase or subsequently the traffic management have to be detected during the planning phase. Therefore, engineers validate the drawings by mentally transferring the plans from 2D visualizations into a 3D space. Augmented Reality (AR) is considered as being able to increase spatial understanding [19] and reduce mental workload [4, 6], since in-situ visualization of information can facilitate mental transfers required to understand the data, as it is commonly discussed, e.g., in the field of wayfinding in AR [12] and utilized in navigation tasks in various large-distance outdoor scenarios [5, 15, 16]. In the context of construction, this advantage is utilized to support workers by visually merging digital information directly into the user's field of view [2, 10]. Here, AR is commonly used, e.g., to visualize data which is not naturally visible (e.g., underground pipes) [19], or which extends the real world (e.g., navigation instructions) [12].



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2281-3/26/04
<https://doi.org/10.1145/3772363.3799371>

In context of the construction phase, AR is considered as having advantages such as performance improvement and error reduction [9]. Considering the construction of roads and highways, AR, explicitly using an AR head-mounted display (HMD), was shown to provide benefits, e.g., by providing a 3D visualization and enabling users to work with digital data hands-free while allowing free movement around the site [8]. In different civil construction contexts, AR is also investigated to be used during the planning phase, e.g., to facilitate the analysis of possible interferences [13], support the communication about plans [17], and allow full-scale walkthrough of plans [14].

As an initial step in an user-centered investigating whether this potential is transferrable to the planning phase of road construction projects, this work presents the development of a prototype to visualize construction drawings in 3D, with a *tabletop* mode to be used in offices, and an *in-situ* mode to be used when visiting the road on-site. Based on this prototype, a preliminary focus group field study in engineering offices and on-site on roads is presented to address the following research questions:

- **RQ1:** Can an AR application support engineers during the road planning phase?
- **RQ2:** Do engineers see potential in using AR to visualize construction drawings?

2 Prototype

A prototype was developed for the Meta Quest 3, using Unity and the Meta XR All-in-One SDK. AR was used through the pass-through mode. As a data basis, actual construction drawings from local engineering firms were obtained, which are currently in use to remodel two roads in a local urban area. Fig. 1a shows a part of one of these drawings. The construction drawings have been created using the computer-aided design (CAD) tool Autodesk Civil 3D [1]. They contain 2D polylines, which represent road elements such as curbs, lane markings, and drainage shafts, as well as 3D objects, e.g., to represent buildings and trees. All data points are represented as georeferenced coordinates. The metadata of the drawings contain additional information such as elevation data. Since for road construction planning 2D information are the most important ones and Unity does not natively support the import of 2D CAD files, these were converted to PNG files beforehand. The construction drawings span a distance of 450m. To efficiently handle the rendering on the Quest 3 while maintaining a great level of detail, the drawings were split into 75m x 75m PNG files, which were rendered next to each other in Unity.

Two visualization modes were developed to show the construction plans: a *tabletop* mode and an *in-situ* mode. The *tabletop* mode mimics the analog use of paper-based construction drawings, as they are spread out on table for inspections and multi-person discussions. The 2D drawings were extended with 3D objects, which were manually modeled as geometric objects and placed on top of the 2D drawings (see Fig. 1b). When being in an engineering office, users can place this virtual drawing on a physical table by defining the table's surface using the Quest 3 controllers. The *in-situ* mode was developed to be used outdoors on the actual road which is to be planned. Here, the construction drawings are overlaid in real-life scale directly onto the user's field of view (see Fig. 1c). At

this stage of the prototype, the positioning is done manually: the drawings are placed so that a predefined geographic position is placed at the Unity coordinate origin and they face north, while in the real-world the user is asked to stand at the same geographic position, also facing north. Manual adjustments can be made using the controllers' joysticks to translate and rotate the drawings.

3 User Study

To evaluate the prototype, a preliminary focus group field study was conducted, where experts used the prototype to assess its usability, usefulness and overall potential to support their workflow. The study was conducted in cooperation with two local engineering firms, taking place both in their offices and outside at the considered roads. For each firm, one focus group was built by inviting employees. For each focus group, the participants were briefed about the study's procedure and purpose, asked to sign a written consent form, and afterwards asked to fill out a demographic questionnaire. Beginning in the offices, the *tabletop* mode was tested first. The experimenter handed the HMD to one of the participants and instructed them to place the virtual construction drawing on a physical table and freely explore it. When the participant decided to be done with the exploration, the HMD was handed to the next participant. After all participants tested the prototype, the experimenter conducted a semi-structured focus group interview, guided by predefined questions regarding their experience and opinion about the prototype. Using the same procedure, the focus group members afterwards also explored the *in-situ* mode. To investigate whether it could even provide potential inside the office by offering a real-life scale 3D view, it was at first used inside the offices and later outside on a road, at its target location. In total, for each focus group the study lasted between 30 and 90 minutes, where each participant was wearing the HMD for up to 10 minutes. In total, 13 experts participated, 7 women and 6 men, aged 27 - 53 ($M = 36$, $SD = 9.31$), with a mean working experience of 13.8 years ($SD = 9.87$). Nine participants described their role as civil engineers, one as road construction engineer, one as construction technician, one as project engineer, and one as engineer. Everyone stated to use a PC during their daily work, while eleven participants also use a tablet, nine use a laptop, and also nine use a smartphone. Eight participants do not have any experience with AR, three use AR less than once a year and two participants use AR at least once a year.

4 Results

The focus group interviews were audio-recorded and later transcribed and analyzed following an inductive thematic approach [3], by coding the transcripts and later grouping the codes using Miro [11]. In the following, the resulting themes will be presented.

4.1 Usefulness of AR to support road construction planning

Overall participants expressed excitement about the technology of AR and its capabilities. In particular, they expect AR to facilitate the detection of errors during the planning phase. Especially using the *in-situ* mode, inaccuracies might be easier to detect compared to when inspecting the plans on a desktop interface. Participants discussed a better understanding of the construction project when

viewed on-site, especially with regard to their own assessment of how visibility conditions, oncoming traffic, and overtaking maneuvers would work, and whether proportions, e.g., for the width of bicycle lanes and safety distances would be maintained. Here, the application could help comparing the numbers in the construction drawings with the individual perception on-site. The *tabletop* mode was mostly considered as helpful to present and discuss plans among stakeholders. In order for the AR application to be actually used, engineers consider simple integration of the system into their workflow to be crucial. For instance, high effort to import construction drawings into AR would drastically reduce the added value.

4.2 Perceived usability, interaction and visualization

Interacting with the application was perceived as uncomplicated, however partly as unintuitive. After receiving instructions on how to use the controllers to place the construction drawings in the real world, no issues in this regard were reported. The visualization of the *tabletop* mode was considered as helpful, especially viewing the plans in 3D instead of on traditional 2D interfaces was perceived as improving the understanding of the planning project. Viewing the *in-situ* mode inside the office was considered as confusing and not helpful, however when taking the HMD outside and using it at the intended location, the visualization was perceived positively. Visibility issues were reported regarding line widths, as too thin lines are difficult to see. Also, elements displayed at a large distance were hard to recognize, which participants explained with a low viewing angle and therefore suboptimal perspective. Both modes have been criticized for their lack of elevation data, which leads to inaccurate visualization on non-flat surfaces such as bridges. Regarding the comfort of the HMD, some participants reported a pressure sensation on the head and would expect headaches with prolonged use due to slight tracking delays. In addition, safety concerns were raised, especially since during the planning phase, the road is still open to traffic, which requires users to be careful when using the *in-situ* mode. Participants therefore suggest that the AR application should always be used in pairs, so that one person without an HMD can pay attention to the traffic and ensure the safety of the person wearing the HMD.

4.3 Further features and use case scenarios

The participants discussed further features which they could imagine to be useful when integrated in the application. For instance, a multi-user feature could enable multiple engineers to view a construction drawing together, with laserpointer functionalities allowing for a joint discussion of the plans. While the 3D visualization was perceived positively, the wish for an additional 2D minimap to facilitate navigation and orientation was expressed. Participants wish to be able to adjust the visualization, e.g. by toggling data layers or change thickness and color of lines. Besides viewing the drawings, the engineers would also like to be able to directly edit the drawings in AR, e.g. by moving or deleting lines. Ideally, these changes would be synchronized with the underlying CAD file to be accessed on other devices. Further advanced features were discussed as well. For instance, simulations could be integrated, e.g.,

of traffic or water flow on the road surface. The AR application could also modify the construction drawings, e.g., by automatically adding missing elements when detected in the real world, or by performing topological measurements. Participants mentioned not only the benefits of the application for themselves, but also potential benefits for others. For instance, construction workers might have an improved orientation on the construction site using AR. Especially during the construction phase, when visual guides such as curbs have been demolished, the corresponding lines still being visible in AR might provide advantages. The participants also considered non-experts such as citizens and stakeholders as potential user groups, who may find it easier to understand the plans when viewed in AR than when reading complex traditional construction drawings and mentally transfer them into 3D. Thus, AR might help to support participation by easily demonstrating construction projects in an understandable way.

5 Discussion

The primary aim of this work was to investigate whether an AR application can support engineers during the planning phase of road construction projects (RQ1), and whether they see potential in using AR for the visualization of construction drawings (RQ2). Overall, the developed application was perceived as being useful and good to use. Both modes, the *tabletop* mode inside offices and the *in-situ* mode on-site at the road, were successfully used by the participants, with only minor usability issues, while they expressed the feeling of increased spatial understanding of the plans, which preliminary indicates that AR could actually support engineers (RQ1). Even though an increased spatial understanding was just reported subjectively, this finding aligns with previous literature which indicates that AR actually can increase spatial understanding and reduce mental workload [4, 6, 19]. RQ2 can also be answered positively, since the participants expressed to see potential of AR, not only for viewing construction drawings, but also for directly editing them in AR or discussing them in multi-user scenarios. They also suggested the extension of the application for other user groups, such as construction workers or citizens. While the *tabletop* mode was mostly discussed as a tool to present and discuss plans with stakeholders, similar on how traditional paper-based plans spread out on a conference table are used, the *in-situ* mode was perceived as having the potential to increase the engineers' spatial understanding, which the participants expect to facilitate the detection of issues and planning errors.

Overall, these findings show that engineers see the potential of AR for the visualization of road construction drawings. Also beyond the sole visualization, they envision an AR application as a multifunctional tool that also supports them in tasks like editing and communicating. This suggests that it is worthwhile to further investigate AR in this field, develop a more sophisticated prototype, and eventually evaluate whether AR can actually support engineers, e.g., in terms of error detection and mental workload.

However, challenges were also reported, which should be considered for future developments of in-situ AR visualization applications. These include headset-specific issues such as slight discomfort when wearing the HMD, with increasing discomfort to be expected during prolonged use, and design problems like reduced

visibility due to thin lines and flat viewing angles of a floor-aligned visualization, but also safety concerns when using the application in traffic situations, which should generally be taken into account for in-situ AR applications. To actually utilize AR, the participants stressed the requirement for an easy and effortless integration of the application into their workflow, such as a fully functional synchronization of the drawings. For instance, for both modes, participants criticized missing elevation data. While for the current prototype, the drawings were converted into PNG files before importing them, which removes metadata including the elevation data, only 2D representation of the plans were visualized. For future prototypes, a full integration should be implemented to import the raw CAD files into the application. Further, participants explicitly stressed that AR visualizations have to be spatially correct when visualized in-situ, which requires a more sophisticated positioning than the manual approach employed so far. With a full file integration, the drawings would keep their georeferenced position information. To actually utilize these information, also the georeferenced position and orientation of the HMD must be determined. This could be achieved by streaming GPS and compass data from a accompanying smartphone, while visual cues can be used to refine accuracy [18]. Also, to realize additional features like editing the drawings in AR, a synchronized integration is mandatory to directly modify the original file. This underscores that integrating AR into an existing workflow to move beyond the status of a naive prototype requires not only a thoughtful design of the application itself, but also an implementation in coordination with existing data streams and interfaces.

6 Conclusion

This work presents a first step in investigating the use of AR to support engineers working with road construction drawings. A prototype was developed to visualize these drawings in 3D, including two modes: a *tabletop* mode and an *in-situ* mode. The prototype was qualitatively evaluated in a preliminary focus group field study involving road engineers. Overall, the results show that engineers see potential in using AR for road construction planning, suggesting that this use case should be investigated further in future work, both in terms of more sophisticated prototypes and more extensive quantitative studies. Besides application specific challenges, the results also present general challenges regarding of using AR in traffic environments for in-situ visualizations, such as safety concerns and the need for a sophisticated software integration, reported from the perspective of potential end users.

Acknowledgments

The authors received funding from the European Union's Horizon Europe research and innovation program for the PRESENCE project (grant agreement No 101135025), the MSCA Staff Exchanges project PLACES (grant agreement No 101086206), the Federal Ministry of Research, Technology and Space (BMFTR) Rescue-Mate (13N16836), and the German Research Foundation (DFG) EYERES (LA 952/11). We acknowledge the cooperation with the local engineering firms and wish to thank the participating engineers in particular.

References

- [1] Autodesk. 2026. Autodesk Civil 3D 2025 | Get Prices & Buy Civil 3D Software. <https://www.autodesk.com/products/civil-3d/overview>.
- [2] Amir H. Behzadan, Suyang Dong, and Vineet R. Kamat. 2015. Augmented Reality Visualization: A Review of Civil Infrastructure System Applications. *Advanced Engineering Informatics* 29, 2 (April 2015), 252–267. doi:10.1016/j.aei.2015.03.005
- [3] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (Jan. 2006), 77–101. doi:10.1191/1478088706qp0630a
- [4] Josef Buchner, Katja Buntins, and Michael Kerres. 2022. The Impact of Augmented Reality on Cognitive Load and Performance: A Systematic Review. *Journal of Computer Assisted Learning* 38, 1 (2022), 285–303. doi:10.1111/jcal.12617
- [5] Julia Hertel, Susanne Schmidt, Marc Briede, Oliver Anders, Thomas Thies, and Frank Steinicke. 2023. Welcome AboARd! Evaluating Augmented Reality as a Skipper's Navigator. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 205–213. doi:10.1109/ISMAR59233.2023.00035
- [6] Nor Farzana Syaza Jeffri and Dayang Rohaya Awang Rambli. 2021. A Review of Augmented Reality Systems and Their Effects on Mental Workload and Task Performance. *Heliyon* 7, 3 (March 2021), e06277. doi:10.1016/j.heliyon.2021.e06277
- [7] Carsten Karcher, Dirk Jansen, Klaus Krass, and Edeltraud Straube. 2024. *Straßenbau und Straßenerhaltung*. Erich Schmidt Verlag GmbH & Co. KG. doi:10.37307/b.978-3-503-23920-7
- [8] Rita El Kassiss, Steven K. Ayer, Mounir El Asmar, and Pingbo Tang. 2023. Discovering Factors That Influence the Use of Augmented Reality for Communication on Active Highway Construction Sites. *Transportation Research Record: Journal of the Transportation Research Board* 2677, 5 (May 2023), 376–389. doi:10.1177/03611981221131311
- [9] Ali Zabihi Kolaei, Erfan Hedayati, Mostafa Khanzadi, and Gholamreza Ghodrati Amiri. 2022. Challenges and Opportunities of Augmented Reality during the Construction Phase. *Automation in Construction* 143 (Nov. 2022), 104586. doi:10.1016/j.autcon.2022.104586
- [10] Jagannath Mallela, Kevin Gilson, and Paul M. Goodrum. 2020. Leveraging Augmented Reality for Highway Construction. <https://rosap.nhtl.bts.gov>.
- [11] Miro. 2026. Der Workspace für KI-Innovation | Miro. <https://miro.com/de/>.
- [12] Zhiwen Qiu, Armin Mostafavi, and Saleh Kalantari. 2025. Use of Augmented Reality in Human Wayfinding: A Systematic Review. *Virtual Reality* 29, 4 (Sept. 2025), 154. doi:10.1007/s10055-025-01226-w
- [13] Moacir Rodrigues. 2021. Civil Construction Planning Using Augmented Reality. In *Sustainability and Automation in Smart Constructions*, Hugo Rodrigues, Florindo Gaspar, Paulo Fernandes, and Artur Mateus (Eds.). Springer International Publishing, Cham, 211–217. doi:10.1007/978-3-030-35533-3_24
- [14] Mukesh Kumar Rohil and Yemini Ashok. 2022. Visualization of Urban Development 3D Layout Plans with Augmented Reality. *Results in Engineering* 14 (June 2022), 100447. doi:10.1016/j.rineng.2022.100447
- [15] Aaron Rowen, Martha Grabowski, Jean-Philippe Rancy, and Alyssa Crane. 2019. Impacts of Wearable Augmented Reality Displays on Operator Performance, Situation Awareness, and Communication in Safety-Critical Systems. *Applied Ergonomics* 80 (Oct. 2019), 17–27. doi:10.1016/j.apergo.2019.04.013
- [16] Javier Santana-Fernández, Jaime Gómez-Gil, and Laura Del-Pozo-San-Cirilo. 2010. Design and Implementation of a GPS Guidance System for Agricultural Tractors Using Augmented Reality Technology. *Sensors* 10, 11 (Nov. 2010), 10435–10447. doi:10.3390/s101110435
- [17] Søren S. Sørensen. 2014. Augmented Reality for Improved Communication of Construction and Maintenance Plans in Nuclear Power Plants. In *Progress of Nuclear Safety for Symbiosis and Sustainability*, Hidekazu Yoshikawa and Zhijian Zhang (Eds.). Springer Japan, Tokyo, 269–274. doi:10.1007/978-4-431-54610-8_27
- [18] Jonas Stehlin, Aaron Häusler, Kordian Caplazi, and Andreas Kunz. 2024. Geospatial Positioning Using a Smartphone and the MS HoloLens. In *Human Systems Engineering and Design (IHSED2024) Future Trends and Applications*. doi:10.54941/ahfe1005540
- [19] Stefanie Zollmann, Raphael Grasset, Gerhard Reitmayr, and Tobias Langlotz. 2014. Image-Based X-ray Visualization Techniques for Spatial Understanding in Outdoor Augmented Reality. In *Proceedings of the 26th Australian Computer-Human Interaction Conference on Designing Futures: The Future of Design*. ACM, Sydney New South Wales Australia, 194–203. doi:10.1145/2686612.2686642